Parametric Azimuth-Variant Motion Compensation for Forward-Looking Multichannel SAR Imagery

Jingyue Lu, Lei Zhang, *Member, IEEE*, Yinghui Quan, *Member, IEEE*, Zhichao Meng, and Yunhe Cao, *Member, IEEE*

***Abstract*— Forward-looking multichannel synthetic aperture radar (FLMC-SAR) is an important tool for modern remote sensing applications, which has the capability to reconstruct the high-resolution image of the front area. However, due to the azimuth-variant characteristics of the motion errors over a long aperture, FLMC-SAR data processing is usually a challenging task, especially when involving the motion compensation (MOCO) coupled with Doppler ambiguity resolving. To accomplish an accurate MOCO for FLMC-SAR, a novel parametric azimuth-variant MOCO approach is proposed in this article. Aiming at the coupling problem of MOCO and Doppler ambiguity resolving over the full aperture, we can decouple them through the subaperture division. As a full synthetic aperture is decomposed into several subapertures, the high-order motion errors of the full aperture can be decomposed into the first-order motion errors of the subaperture. On this basis, the mismatch of the space–time spectrum caused by the motion errors can be solved by spectral estimation, yielding Doppler ambiguity resolving for each subaperture. Meanwhile, the azimuth-variant characteristic of motion errors in FLMC-SAR system is characterized by a parametric angle-dependent quadratic phase error (QPE) model. The motion parameters are estimated by a joint multichannel angle estimation-based signal quadratic decomposition method. Immediately, the MOCO for ambiguous targets with different motion errors can be processed separately to improve the imaging performance. Experimental results based on both simulated and real data demonstrate that the proposed method is suitable for FLMC-SAR system.**

***Index Terms*— Azimuth variant, Doppler ambiguity, forward-looking multichannel synthetic aperture radar (FLMC-SAR), motion compensation (MOCO), signal quadratic decomposition, subaperture.**

I. Introduction

SYNTHETIC aperture radar [1]–[6] (SAR) is widely used in practical engineering to generate high-resolution microwave images. With appropriate geometry configurations, SAR is capable of reconstructing the high-resolution image of the front area, which we call forward-looking SAR. Since forward-looking SAR has all-weather, day, and night forward-looking imaging features that greatly improve the radar's ability to acquire information, it is widely used in engineering applications, such as aircrafts and civilian remote sensing. The earliest forward-looking SAR system was proposed by the German Aerospace Center, called Sector Imaging Radar for Enhanced Vision (SIREV) [7], [8], and some related research works, such as chirp scaling algorithm [9], compressive sensing algorithm [10], and 3-D imaging algorithm [11], [12], are suitable for the conventional SIREV system. Bistatic forward-looking synthetic aperture radar (BFSAR) [13], [14] is also an important system to realize the forward-looking imaging. In [15]–[17], some motion compensation (MOCO) and autofocus algorithms are proposed to improve the imaging quality of BFSAR. Moreover, without the requirement of target relative motion, radar coincidence imaging (RCI) [18], [19] has shown great potential in forward-looking imaging. Another important tool of high-resolution forward-looking imaging is based on multichannel SAR [20]–[25] system, which we call forward-looking multichannel SAR [26], [27] (FLMC-SAR). However, in practical applications, due to the atmospheric turbulence, the radar platform often probably deviates from the ideal linear trajectory. Those deviations as well as variations in platform velocity lead to the problem of motion errors, which in turn severely degrade the imaging quality [28], [29]. Such motion errors can be measured by the inertial navigation system (INS)/global positioning system (GPS) [30] and estimated by raw data. However, based on the data measured by INS/GPS, the MOCO does not prove sufficiently accurate for high-quality imaging. Therefore, the INS/GPS system output is often used to achieve a coarse MOCO, such as the first-order azimuth-invariant MOCO. Then, the residue motion errors can be obtained through autofocus [31]–[35] to achieve more precise MOCO [36]. For this reason, MOCO based on the raw data is crucial for FLMC-SAR systems.

In recent years, the raw data-based MOCO has been well explored. Phase gradient autofocus (PGA) [35], one of the most popular raw data-based MOCO algorithms, proves very robust to a variety of imagery and phase error functions in different imaging modes [37], [38]. In addition, some extended PGA algorithms have been developed to improve the accuracy and robustness of phase estimation, including the quality PGA (QPGA) [39] and the weighted PGA (WPGA) [40], [41]. In [42], the range-dependent phase errors are removed by

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a modified WPGA algorithm using the weighting and local maximum-likelihood (LML) estimate. Other strategies based on the Doppler rate estimation are proposed to perform MOCO via the map-drift algorithm (MDA) [43]–[47] and also found to demonstrate strong robustness. By minimizing the misregistration caused by the Doppler frequency modulation rate errors in the azimuth matched filter, the parameters of the matched filter are determined. In [47], the phase error function is modeled as a linear function of the azimuth coordinate before its parameters are estimated via taking the maximum of the integral on the scaled correlation map. Unfortunately, neither MDA nor PGA can be directly applied in FLMC-SAR imaging because there are several issues that should be considered when it comes to FLMC-SAR MOCO.

1) Doppler ambiguity makes the different motion errors of the ambiguous targets aliased in the same bin, which increases the complexity of MOCO. Moreover, the motion errors cause the mismatch of the space–time spectrum, seriously decreasing the performance of Doppler ambiguity resolving. Therefore, whether we intend to perform MOCO or resolve Doppler ambiguity, we have to think about both. This coupled problem of MOCO and Doppler ambiguity resolution increases the challenge of imaging. 2) Under the FLMC-SAR imaging mode, the phase errors caused by the cross-track motion errors in the left and right scenes prove opposite to each other. Thus, the MOCO for the left and right scenes should be implemented. 3) The variation of motion errors with the azimuth should be considered carefully in FLMC-SAR MOCO because the mismatch of space–time spectrum caused by these motion errors also becomes azimuth-variant.

To cope with those problems, a novel parametric azimuth-variant MOCO approach is proposed in this article. First, the motion errors are approximately invariant within a short azimuth aperture. In this proposed method, we decompose a full synthetic aperture into several subapertures so that the high-order motion errors of the full aperture can then be decomposed into the first-order motion errors of the subapertures. Thus, we can separate the two coupled problems of MOCO and Doppler ambiguity resolving and deal with them individually within each subaperture. Second, because of this subaperture division, the mismatch of space–time spectrum caused by the motion errors can be regarded as azimuth-invariant. Consequently, we can utilize spectral estimation to solve the mismatch of the space–time spectrum. Third, due to the limited azimuth resolution in each subaperture, the influence of higher order phase errors on the imaging resolution is negligible. Therefore, a parametric quadratic phase error (QPE) model can be established to characterize the phase errors. Through this model, the azimuth-variant phase errors are modeled as an angle-dependent function. In this case, the angle estimation can be achieved through the degrees of freedom of the multichannel. As a result, a joint multichannel angle estimation-based signal quadratic decomposition method is proposed to estimate the motion parameters of the QPE model. In this QPE model, motion errors are divided into two items. Before Doppler ambiguity resolving, the angle absolute value-dependent azimuth-variant phase errors for the left and right scenes can be compensated, but the angle-dependent azimuth-variant phase errors should be compensated individually for the left and right scenes after Doppler ambiguity resolving. Finally, the subaperture data after the space–time spectrum correction and MOCO are coherently combined to recover the full-aperture data, yielding high-resolution FLMC-SAR imagery. The main innovations of the proposed method are as follows.

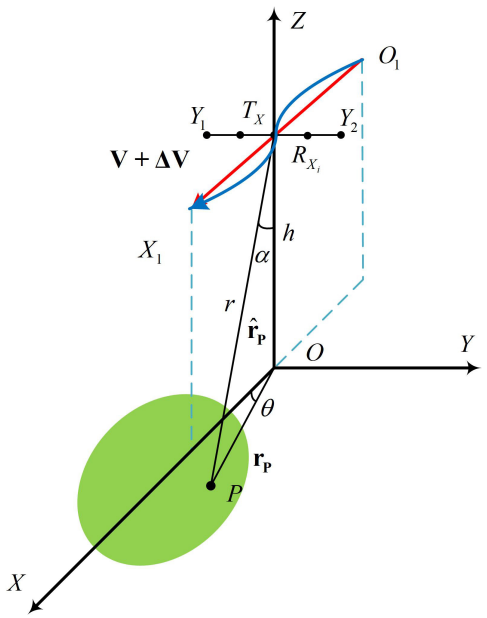

Fig. 1. Geometry model of spotlight FLMC-SAR.

1) To cope with the coupling between MOCO and Doppler ambiguity resolving, a full synthetic aperture is decomposed into several subapertures; immediately, the high-order motion errors of the full aperture can be decomposed into the first-order motion errors of the subapertures. On this basis, the space–time spectrum correction is achieved by spectral estimation. Therefore, the MOCO and Doppler ambiguity resolving can be performed individually.
2) In order to precisely perform the azimuth-variant MOCO, a parametric QPE model is established to characterize the phase errors. Then, a joint multichannel angle estimation-based signal quadratic decomposition method is proposed to estimate the motion parameters. After that, the MOCO for ambiguous targets with different motion errors can be processed separately.

This article is organized as follows. Section II gives the signal and geometry model of FLMC-SAR. Section III presents an FLMC-SAR imaging method, and the Doppler ambiguity in FLMC-SAR imaging is analyzed in detail. Section IV analyzes the motion errors in FLMC-SAR, and a corresponding parametric azimuth-variant MOCO is proposed. The experimental results based on both simulated and real data are given in Section V. Section VI concludes this work.

II. FLMC-SAR System Model

The geometry model of spotlight FLMC-SAR is shown in Fig. 1. The radar system adopts the array antenna with

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multiple channels. This antenna is symbolized as the line segment Y_1Y_2, which is paralleled with the Y-axis in the YZ plane. The antenna channels are symmetrically distributed on Y_1Y_2. d is the distance between the adjacent channels. Each channel of the array antenna can receive signals. The ith receiving channel is denoted by R_{X_i}. Besides, the channel at the center of the array can transmit signals as well, denoted as T_X.

In an ideal case, the array antenna is assumed to move along the X-axis with the same velocity $\mathbf{V} = [v, 0, 0]$ and the nominal trajectory is the straight line O_1X_1. However, due to the atmospheric turbulence and so on, the velocity deviations $\Delta\mathbf{V} = [\Delta v_x, \Delta v_y, \Delta v_z]$ are caused. Thus, the actual trajectory becomes a curve line deviating from the nominal trajectory.

The symbol P denotes an arbitrary target located at imaging area. The range between the target and the antenna phase center (APC) is the reference slant range r of the target. We define the vector pointing toward the target from the APC as the radar light of sight $\hat{\mathbf{r}}_\mathbf{P}[r\sin\alpha\cos\theta, r\sin\alpha\sin\theta, -r\cos\alpha]$. The angle between the radar light of sight and the Z-axis is the beam pointing pitch angle α. The angle between the projection of the radar light of sight on the ground (ie. XY-plane) and the X-axis is the beam pointing azimuth angle θ. Then, the position of the point target P is $\mathbf{r_P}[r\sin\alpha\cos\theta, r\sin\alpha\sin\theta, 0]$. The symbol t_m represents the slow time. It is zero when the APC of the radar is located at the center of the synthetic aperture. The actual APC position at the slow time t_m is $\mathbf{R_X}[(v+\Delta v_x)t_m, \Delta v_y t_m, \Delta v_z t_m + r cos\alpha]$. Then, the instantaneous range is given by (1), as shown at the bottom of the next page. The instantaneous range can be divided into two terms. The first term of (2) is the straight trajectory without the deviations, and the second term of (2), as shown at the bottom of the next page is the instantaneous range error, which is caused by the velocity deviations $\Delta\mathbf{V} = [\Delta v_x, \Delta v_y, \Delta v_z]$.

Assume that the linear frequency-modulated (LFM) signal with chirp rate γ and wavelength λ is transmitted. Through the downconversion to the baseband, the received signal from target P can be expressed by

$$\begin{aligned} s_P(\hat{t}, t_m; r, \theta) = {} & A_P \cdot w_r\left(\hat{t} - \frac{2R(t_m; r, \theta)}{c}\right) \\ & \cdot \exp\left[j\pi\gamma\left(\hat{t} - \frac{2R(t_m; r, \theta)}{c}\right)^2\right] \\ & \cdot \exp\left[-j\frac{4\pi}{\lambda}R_0(t_m; r, \theta)\right] \\ & \cdot \exp\left[-j\frac{4\pi}{\lambda}\Delta R_0(t_m; r, \theta, \Delta v_x, \Delta v_y, \Delta v_z)\right] \end{aligned} \tag{5}$$

where A_P is the complex reflectivity of target P, w_r is the window function in range, c is the light speed, and $\hat{t}$ is the fast time.

III. FLMC-SAR Imaging

In order to better present our FLMC-SAR MOCO method, let us first introduce the related FLMC-SAR imaging work.

A. Reference Target Filtering

Subjected to range pulse compression processing, the signal can be expressed as

$$\begin{aligned} & s_{rc}(\hat{t}, t_m; r, \theta) \\ & = \mathrm{IFFT}_{\hat{t}}\{\mathrm{FFT}_{\hat{t}}[s_P(\hat{t}, t_m; r, \theta)] \cdot \mathrm{FFT}_{\hat{t}}[s_{\mathrm{ref}}(\hat{t})]\} \\ & = A_P \cdot \mathrm{sinc}\left[B(\hat{t} - \frac{2R(t_m; r, \theta)}{c})\right] \\ & \quad \cdot \exp\left[-j\frac{4\pi}{\lambda}R_0(t_m; r, \theta)\right] \\ & \quad \cdot \exp\left[-j\frac{4\pi}{\lambda}\Delta R_0(t_m; r, \theta, \Delta v_x, \Delta v_y, \Delta v_z)\right] \end{aligned} \tag{6}$$

where B is the signal bandwidth, $\mathrm{FFT}_{\hat{t}}[\cdot]$ is the range Fourier transform operator, $\mathrm{IFFT}_{\hat{t}}[\cdot]$ is the range inverse Fourier transform operator, $\mathrm{sinc}(\hat{t})$ is the sampling function, and $s_{\mathrm{ref}}(\hat{t})$ is the matching filter function, which can be described by

$$s_{\mathrm{ref}}(\hat{t}) = w_r(\hat{t}) \cdot \exp(-j\pi\gamma\hat{t}^2) \tag{7}$$

where w_r is the window function in range. By transforming (6) into the range frequency domain via a range Fourier transform, we obtain

$$\begin{aligned} & S_{rc}(f_r, t_m; r, \theta) \\ & = \mathrm{FFT}_{\hat{t}}[s_{rc}(\hat{t}, t_m; r, \theta)] \\ & = A_P \cdot S(f) \cdot \exp\left[-j4\pi(f_c + f_r)\frac{R(t_m; r, \theta)}{c}\right] \\ & = A_P \cdot S(f) \cdot \exp\left[-j4\pi(f_c + f_r)\frac{R_0(t_m; r, \theta)}{c}\right] \\ & \quad \cdot \exp\left[-j4\pi(f_c + f_r)\frac{\Delta R(t_m; r, \theta, \Delta v_x, \Delta v_y, \Delta v_z)}{c}\right] \end{aligned} \tag{8}$$

where $S(f) = \mathrm{FFT}_{\hat{t}}[\mathrm{sinc}(B\hat{t})]$ and f_c is the carrier frequency. Suppose that the point target with the beam pointing azimuth angle 0° and the reference slant range r_0 is selected as the reference target in the center of the scene. Then, the reference target filtering can be expressed as

$$\begin{aligned} & S_1(f_r, t_m; r, \theta) \\ & = S_{rc}(f_r, t_m; r, \theta) \cdot \exp\left[j4\pi(f_c + f_r)\frac{R_0(t_m; r_0, 0)}{c}\right] \\ & = A_P \cdot S(f) \\ & \quad \cdot \exp\left\{-j4\pi(f_c + f_r)\frac{[R(t_m; r, \theta) - R_0(t_m; r_0, 0)]}{c}\right\} \\ & = A_P \cdot S(f) \\ & \quad \cdot \exp\left\{-j4\pi(f_c + f_r)\frac{[R_0(t_m; r, \theta) - R_0(t_m; r_0, 0)]}{c}\right\} \\ & \quad \cdot \exp\left[-j4\pi(f_c + f_r)\frac{\Delta R(t_m; r, \theta, \Delta v_x, \Delta v_y, \Delta v_z)}{c}\right]. \end{aligned} \tag{9}$$

Applying the range inverse Fourier transform to (9), we can obtain the following reference target filtered signal:

$$\begin{aligned} & s_1(\hat{t}, t_m; r, \theta) \\ & = \mathrm{IFFT}_{\hat{t}}[S_1(f_r, t_m; r, \theta)] \\ & = A_P \cdot \mathrm{sinc}\left[B(\hat{t} - \frac{2(R(t_m; r, \theta) - R_0(r_0, 0))}{c})\right] \end{aligned}$$

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$$\cdot \exp\left\{-j\frac{4\pi}{\lambda}[R_0(t_m;r,\theta)-R_0(t_m;r_0,0)]\right\}$$
$$\cdot \exp\left[-j\frac{4\pi}{\lambda}\Delta R(t_m;r,\theta,\Delta v_x,\Delta v_y,\Delta v_z)\right]. \quad (10)$$

The Taylor series of $R_0(t_m;r,\theta)-R_0(t_m;r_0,0)$ can be expressed as

$$R_0(t_m;r,\theta)-R_0(t_m;r_0,0)$$
$$\approx r-r_0-\sin\alpha v(\cos\theta-1)t_m$$
$$+\left(\frac{2v^2-\sin^2\alpha v^2\cos^2\theta}{2r}-\frac{2v^2-\sin^2\alpha v^2}{2r_0}\right)t_m^2. \quad (11)$$

Then, (10) can be approximately rewritten as

$$s_1(\hat{t},t_m;r,\theta)$$
$$\approx A_P\cdot \mathrm{sinc}\left[B(\hat{t}-\frac{2(r-r_0)}{c})]\cdot\exp[-j\frac{4\pi}{\lambda}(r-r_0)\right]$$
$$\cdot \exp\left\{-j\frac{4\pi}{\lambda}[-\sin\alpha v(\cos\theta-1)t_m]\right\}$$
$$\cdot \exp\left\{-j\frac{4\pi}{\lambda}\left[\left(\frac{2v^2-\sin^2\alpha v^2\cos^2\theta}{2r}-\frac{2v^2-\sin^2\alpha v^2}{2r_0}\right)t_m^2\right]\right\}$$
$$\cdot \exp\left[-j\frac{4\pi}{\lambda}\Delta R(t_m;r,\theta,\Delta v_x,\Delta v_y,\Delta v_z)\right]. \quad (12)$$

Equation (12) can be viewed as consisting of five terms. The first term of (12) $A_P\cdot\mathrm{sinc}[B(\hat{t}-(2(r-r_0)/c))]$ is the signal envelope in range. The second term of (12) $\exp[-j(4\pi/\lambda)(r-r_0)]$ is a complex constant related to the reference target. The third term of (12) $\exp\{-j(4\pi/\lambda)[-\sin\alpha v(\cos\theta-1)t_m]\}$ is the Doppler frequency phase, which determines the azimuth coordinate of the target in the imaging result. The fourth term of (12) $\exp\{-j(4\pi/\lambda)[((2v^2-\sin^2\alpha v^2\cos^2\theta/2r)-(2v^2-\sin^2\alpha v^2/2r_0))t_m^2]\}$ is the angle absolute value-dependent azimuth-variant residual phase errors. Also, the fifth term of (12) $\exp[-j(4\pi/\lambda)\Delta R(t_m;r,\theta,\Delta v_x,\Delta v_y,\Delta v_z)]$ is the angle-dependent azimuth-variant motion phase errors.

B. Left–Right Doppler Ambiguity in FLMC-SAR

Left–right Doppler ambiguity is an inherent problem in forward-looking imaging. In this geometric model of the forward-looking mode, the two point targets that are symmetrical about the trajectory have the same slant range history, which results in the left–right Doppler ambiguity in FLMC-SAR.

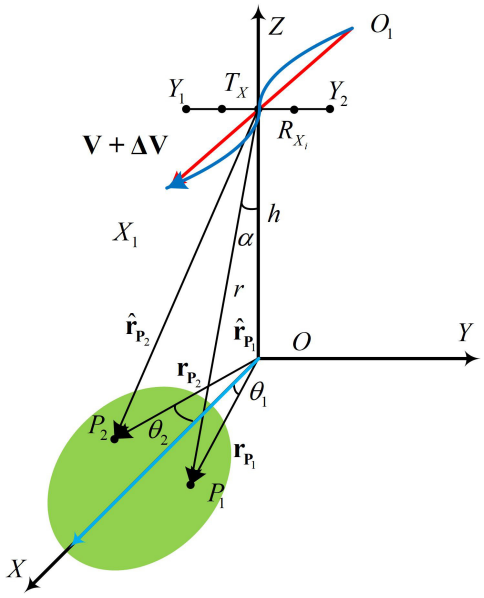

Fig. 2. Geometry model with two symmetric targets.

As shown in Fig. 2, P_1 and P_2 are the two point targets with the same reference slant range, and their positions can be expressed as $\mathbf{r_{P}}_1[r\sin\alpha\cos\theta_1, r\sin\alpha\sin\theta_1, 0]$ and $\mathbf{r_{P}}_2[r\sin\alpha\cos\theta_2, r\sin\alpha\sin\theta_2, 0]$, respectively. Also, the slant range difference between P_1 and P_2 is shown in (13), as shown at the bottom of the page 6.

In the case where the two point targets that are symmetrical about the trajectory share the same Doppler history, the echo signals of the symmetric targets will be accumulated in the same Doppler bin, causing the left–right Doppler ambiguity of FLMC-SAR. The same Doppler history of the targets can be represented by the same slant range history of the targets. By setting (13) to zero, the left–right Doppler ambiguity relationship can be obtained. According to (14), as shown at the bottom of the page 6, the center of the symmetry angle can be expressed as follows.

In the case where the two point targets that are symmetrical about the trajectory share the same Doppler history, the echo signals of the symmetric targets will be accumulated in the same Doppler bin, causing the left–right Doppler ambiguity of FLMC-SAR. The same Doppler history of the targets can be represented by the same slant range history of the targets.

$$\frac{\sin\theta_1+\sin\theta_2}{2}=\frac{\Delta v_y t_m}{r\sin\alpha}. \quad (15)$$

$$R_R(t_m;r,\theta)=|\mathbf{R_X}-\mathbf{r_P}|$$
$$=\sqrt{((v+\Delta v_x)t_m-r\sin\alpha\cos\theta)^2+(\Delta v_y t_m-r\sin\alpha\sin\theta)^2+(\Delta v_z t_m+r\cos\alpha)^2} \quad (1)$$
$$R(t_m;r,\theta)=\sqrt{(vt_m+\Delta v_x t_m-r\sin\alpha\cos\theta)^2+(\Delta v_y t_m-r\sin\alpha\sin\theta)^2+(\Delta v_z t_m+r\cos\alpha)^2}$$
$$\approx R_0(t_m;r,\theta)+\Delta R_0(t_m;r,\theta,\Delta v_x,\Delta v_y,\Delta v_z) \quad (2)$$
$$R_0(t_m;r,\theta)=\sqrt{(vt_m-r\sin\alpha\cos\theta)^2+(r\sin\alpha\sin\theta)^2+(r\cos\alpha)^2} \quad (3)$$
$$\Delta R_0(t_m;r,\theta,\Delta v_x,\Delta v_y,\Delta v_z)=\frac{vt_m-r\sin\alpha\cos\theta}{R_0(t_m;r,\theta)}\Delta v_x t_m-\frac{r\sin\alpha\sin\theta}{R_0(t_m;r,\theta)}\Delta v_y t_m+\frac{r\cos\alpha}{R_0(t_m;r,\theta)}\Delta v_z t_m \quad (4)$$

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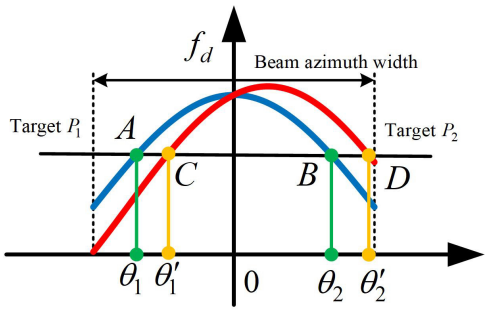

Fig. 3. Space–time spectrum of ambiguous targets.

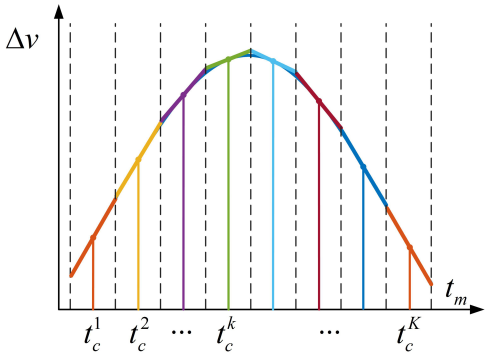

Fig. 4. Approximate estimation result of velocity errors.

Equation (15) shows that the velocity error Δv_y will affect the Doppler ambiguity relationship. For simplicity, we only consider the effect of Δv_y on the Doppler ambiguity relationship. In the absence of the velocity error Δv_y, (15) can be expressed as

$$\frac{\sin\theta_1+\sin\theta_2}{2}=0. \tag{16}$$

Also, the Doppler frequency can be calculated as

$$f_d=\frac{2\mathbf{V}\cdot\hat{\mathbf{r}}_\mathbf{P}}{\lambda\cdot|\hat{\mathbf{r}}_\mathbf{P}|}=\frac{2v\sin\alpha\cos\theta}{\lambda}. \tag{17}$$

Otherwise, the Doppler ambiguity relationship is still expressed as (15), and the Doppler frequency can be denoted as

$$\begin{aligned} f_d &= \frac{2(\mathbf{V}+\Delta\mathbf{V})\cdot\hat{\mathbf{r}}_\mathbf{P}}{\lambda\cdot|\hat{\mathbf{r}}_\mathbf{P}|} \\ &= \frac{2(v\sin\alpha\cos\theta+\Delta v_y\sin\alpha\sin\theta)}{\lambda}. \end{aligned} \tag{18}$$

According to (17) and (18), the space–time spectrum between the azimuth angle and the Doppler frequency can be obtained, as shown in Fig. 3. The blue curve represents the space–time spectrum without motion errors, but the red curve represents the space–time spectrum with motion error Δv_y. Assume that the two point targets P_1 and P_2 have the same Doppler frequency f_d, and thus, the two targets have left–right Doppler ambiguity. If there is no motion errors, a straight line F perpendicular to the f_d-axis intersects the blue curve at points A and B. The azimuth angle coordinates of the two points, θ_1 and θ_2, respectively, denote the azimuth angles of the two point targets P_1 and P_2, satisfying (16). However, if there is motion error Δv_y, the straight line F intersects the red curve at points C and D. The azimuth angles of the two point targets P_1 and P_2 correspond to the azimuth angle coordinates of the two intersection points θ_1' and θ_2', respectively. θ_1' and θ_2' satisfy (15). By this comparison, it is demonstrated that the Doppler ambiguity relationship changes with the motion error Δv_y. Moreover, Doppler ambiguity makes the different motion errors of the ambiguous targets aliased in the same bin. Therefore, whether we intend to perform MOCO or resolve Doppler ambiguity, we have to think about both. For avoiding this difficulty, we proposed a method to take these two coupled problems apart and deal with them individually.

IV. Motion Compensation

In this section, we present a robust parametric azimuth-variant MOCO approach, in which the phase errors are composed of two terms. This approach consists of five steps to implement MOCO.

A. Subaperture Division

We decompose a full aperture into K subapertures, each of which is composed of N azimuth pulses. Based on this, we can do the following works.

First, the velocity errors are approximately regarded as a linear function in terms of the slow time t_m in each subaperture. Therefore, the velocity errors are approximately expressed as follows:

$$\Delta v_x = a_x t_m + \Delta v_{x0} \tag{19}$$

$$\Delta v_y = a_y t_m + \Delta v_{y0} \tag{20}$$

$$\Delta v_z = a_z t_m + \Delta v_{z0} \tag{21}$$

where Δv_{x0}, Δv_{y0}, and Δv_{z0} denote the constant velocity bias in the x-, y-, and z-directions, respectively, and a_x, a_y, and a_z denote the acceleration in the x-, y-, and z-directions, respectively. As shown in Fig. 4, the high-order velocity error in the full aperture is decomposed into the primary velocity error in the subaperture.

Second, the Doppler ambiguity relationship can be regarded as approximately invariant in each subaperture. It requires that the slant range difference of the symmetric targets caused by the motion error is small enough to avoid the range cell migration. In this way, the echo signals of the symmetric targets will be accumulated in the same range bin so that the Doppler ambiguity relationship remains the same. Here, the difference is required to be less than a quarter of the range resolution, which is expressed as follows:

$$\begin{aligned} &R(t_m;r,\theta_1)-R(t_m;r,\theta_2) \\ &=r\sin\alpha(\sin\theta_1-\sin\theta_2)\cdot(r\sin\alpha(\sin\theta_1+\sin\theta_2)-2\Delta v_y t_m) \\ &\le\frac{\Delta R}{4}. \end{aligned} \tag{22}$$

As shown in Fig. 5, the variant Doppler ambiguity relationship in the full aperture is approximated as the invariant Doppler ambiguity relationship in the subaperture.

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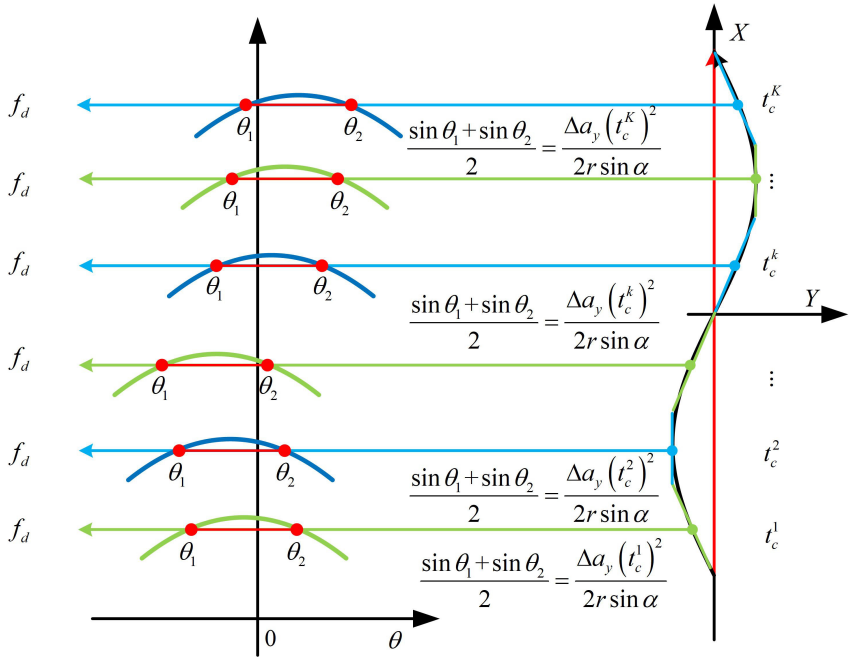

Fig. 5. Doppler ambiguity relationship approximation of ambiguous targets in each subaperture.

Third, due to the mechanical inertia of platform, the along-track velocity varies slowly. Compensated by azimuth resample processing, the phase errors caused by the along-track acceleration a_x is neglected in each subaperture. The instantaneous range error (4) as shown at the bottom of page 4, is approximated by

$$\begin{aligned}&\Delta R_0(t_m; r, \theta, \Delta v_x, \Delta v_y, \Delta v_z)\\ &\approx \frac{v t_m - r\sin\alpha\cos\theta}{R_0(t_m; r, \theta)}(\Delta v_{x0} t_m)\\ &\quad - \frac{r\sin\alpha\sin\theta}{R_0(t_m; r, \theta)}\left(a_y t_m^2 + \Delta v_{y0} t_m\right)\\ &\quad + \frac{r\cos\alpha}{R_0(t_m; r, \theta)}\left(a_z t_m^2 + \Delta v_{z0} t_m\right).\end{aligned} \tag{23}$$

Fourth, in instantaneous range error (4), the instantaneous range is approximately the reference slant range in each subaperture. The instantaneous range error (23) is approximated by

$$\begin{aligned}&\Delta R_0(t_m; r, \theta, \Delta v_x, \Delta v_y, \Delta v_z)\\ &= \frac{v\Delta v_{x0}}{r} t_m^2 - \sin\alpha\sin\theta a_y t_m^2 + \cos\alpha a_z t_m^2\\ &\quad - \sin\alpha\cos\theta\Delta v_{x0} t_m - \sin\alpha\sin\theta\Delta v_{y0} t_m + \cos\alpha\Delta v_{z0} t_m.\end{aligned} \tag{24}$$

Then, the reference target filtered signal within the kth subaperture can be expressed as

$$\begin{aligned}&s_1(\hat{t}, t_m; r, \theta)\\ &\approx A_P\cdot\operatorname{sinc}\left[B\left(\hat{t} - \frac{2(r-r_0)}{c}\right)\right]\cdot\exp\left[-j\frac{4\pi}{\lambda}(r-r_0)\right]\\ &\quad\cdot\exp\left\{-j\frac{4\pi}{\lambda}[-\sin\alpha v(\cos\theta - 1)t_m]\right\}\\ &\quad\cdot\exp\left\{-j\frac{4\pi}{\lambda}(-\sin\alpha\cos\theta\Delta v_{x0}\right.\\ &\qquad\left.-\sin\alpha\sin\theta\Delta v_{y0} + \cos\alpha\Delta v_{z0})t_m\right\}\\ &\quad\cdot\exp\left\{-j\frac{4\pi}{\lambda}\left[\left(\frac{2v^2 - \sin^2\alpha v^2\cos^2\theta}{2r}\right.\right.\right.\\ &\qquad\left.\left.\left.-\frac{2v^2 - \sin^2\alpha v^2}{2r_0}\right)t_m^2\right]\right\}\\ &\quad\cdot\exp\left[j\frac{4\pi}{\lambda}\sin\alpha\sin\theta a_y t_m^2\right]\\ &\quad\cdot\exp\left[-j\frac{4\pi}{\lambda}\left(\frac{v\Delta v_{x0}}{r} + \cos\alpha a_z\right)t_m^2\right]\end{aligned}$$

$$\begin{aligned}R(t_m; r, \theta_1) - R(t_m; r, \theta_2) &= \sqrt{((v+\Delta v_x)t_m - r\sin\alpha\cos\theta_1)^2 + (\Delta v_y t_m - r\sin\alpha\sin\theta_1)^2 + (\Delta v_z t_m - r\cos\alpha)^2}\\ &\quad - \sqrt{((v+\Delta v_x)t_m - r\sin\alpha\cos\theta_2)^2 + (\Delta v_y t_m - r\sin\alpha\sin\theta_2)^2 + (\Delta v_z t_m - r\cos\alpha)^2}\\ &= r^2\sin^2\alpha(\sin^2\theta_1 - \sin^2\theta_2) - 2\Delta v_y t_m r\sin\alpha(\sin\theta_1 - \sin\theta_2)\\ &= r\sin\alpha(\sin\theta_1 - \sin\theta_2)\cdot(r\sin\alpha(\sin\theta_1 + \sin\theta_2) - 2\Delta v_y t_m)\end{aligned} \tag{13}$$

$$r\sin\alpha(\sin\theta_1 - \sin\theta_2)\cdot(r\sin\alpha(\sin\theta_1 + \sin\theta_2) - 2\Delta v_y t_m) = 0 \tag{14}$$

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$$
\begin{aligned}
&= A_P \cdot \operatorname{sinc}\left[B(\hat{t} - \frac{2(r-r_0)}{c})\right] \cdot \exp\left[-j\frac{4\pi}{\lambda}(r-r_0)\right] \\
&\quad \cdot \exp\left\{-j\frac{4\pi}{\lambda}[-\sin\alpha v(\cos\theta - 1)t_m]\right\} \\
&\quad \cdot \exp\{-j\varphi_{\Delta f_d} t_m\} \\
&\quad \cdot \exp\left[-j\frac{4\pi}{\lambda}\gamma_1(\theta)t_m^2\right] \cdot \exp\left[-j\frac{4\pi}{\lambda}\gamma_2(\theta)t_m^2\right] \\
&\quad \cdot \exp\left[-j\frac{4\pi}{\lambda}\gamma_3 t_m^2\right] \\
\varphi_{\Delta f_d} &= \frac{4\pi}{\lambda}(-\sin\alpha\cos\theta\Delta v_{x0} - \sin\alpha\sin\theta\Delta v_{y0} + \cos\alpha\Delta v_{z0}) \\
\gamma_1(\theta) &= \left(\frac{2v^2 - \sin^2\alpha v^2\cos^2\theta}{2r} - \frac{2v^2 - \sin^2\alpha v^2}{2r_0}\right) \\
\gamma_2(\theta) &= -\sin\alpha\sin\theta a_y \\
\gamma_3 &= \frac{v\Delta v_{x0}}{r} + \cos\alpha a_z. \qquad (25)
\end{aligned}
$$

As represented in (25), an azimuth-variant QPE model is established in each subaperture. The chirp rate $\gamma_1(\theta)$ is the angle absolute value-dependent azimuth-variant phase error. The chirp rate $\gamma_2(\theta)$ is the angle-dependent azimuth-variant phase error. The chirp rate γ_3 is the angle-independent azimuth-variant phase error. However, in our algorithm, the chirp rate γ_3 can be estimated and compensated together with $\gamma_2(\theta)$. For simplicity, we classify $\gamma_2(\theta)$ and γ_3 as the same type of phase error, that is, angle-dependent azimuth-variant phase error. As a result, phase errors are divided into two items. Before Doppler ambiguity resolving, the angle absolute value-dependent azimuth-variant phase error $\gamma_1(\theta)$ for the left and right scenes can be compensated, but the angle-dependent azimuth-variant phase errors $\gamma_2(\theta)$ and γ_3 should be compensated individually for the left and right scenes after Doppler ambiguity resolving.

B. Compensation of Angle Absolute Value-Dependent Azimuth-Variant Phase Errors

With the known parameters (v, α, r, and r_0), the mapping relationship between azimuth angle θ and chirp rate $\gamma_1(\theta)$ at the center of each subaperture can be obtained. Then, the phase correction function to remove the angle absolute value-dependent azimuth-variant phase errors is given by

$$
G_{\gamma 1}(\theta) = \exp\left(j\frac{4\pi}{\lambda}\gamma_1(\theta)\left(t_c^k\right)^2\right) \qquad (26)
$$

where t_c^k is the slow time when the APC of the radar is located at the center of the kth subaperture. After the compensation of angle absolute value-dependent azimuth-variant phase errors, the signal becomes

$$
\begin{aligned}
s_2(\hat{t}, t_m; r, \theta) &= \text{IFFT}[\text{FFT}[s_1(\hat{t}, t_m; r, \theta)] * G_{\gamma 1}(\theta)] \\
&= A_P \cdot \operatorname{sinc}\left[B(\hat{t} - \frac{2(r-r_0)}{c})\right] \\
&\quad \cdot \exp\left[-j\frac{4\pi}{\lambda}(r-r_0)\right] \\
&\quad \cdot \exp\{-j\frac{4\pi}{\lambda}[-\sin\alpha v(\cos\theta - 1)t_m]\} \\
&\quad \cdot \exp\{-j\varphi_{\Delta f_d} t_m\} \\
&\quad \cdot \exp\left[-j\frac{4\pi}{\lambda}\gamma_2(\theta)t_m^2\right] \cdot \exp\left[-j\frac{4\pi}{\lambda}\gamma_3 t_m^2\right] \\
\varphi_{\Delta f_d} &= \frac{4\pi}{\lambda}(-\sin\alpha\cos\theta\Delta v_{x0} - \sin\alpha\sin\theta\Delta v_{y0} \\
&\qquad + \cos\alpha\Delta v_{z0}) \\
\gamma_2(\theta) &= -\sin\alpha\sin\theta a_y \\
\gamma_3 &= \frac{v\Delta v_{x0}}{r} + \cos\alpha a_z. \qquad (27)
\end{aligned}
$$

As mentioned above, the angle absolute value-dependent azimuth-variant phase errors for the left and right scenes can be compensated before Doppler ambiguity resolving because this phase error is consistent for the left and right scenes. However, the angle-dependent azimuth-variant phase errors should be compensated individually for the left and right scenes after Doppler ambiguity resolving.

C. Left–Right Doppler Ambiguity Resolving via Beamforming

In this section, we will introduce the Doppler ambiguity resolving. Consider the multiply channels of the system, and the array steering vector is given by (28), as shown at the bottom of the next page where L is the number of array elements, θ is the beam pointing azimuth angle of the target, α is the beam pointing pitch angle, and d is the distance of the adjacent array elements.

In the case of two symmetrical targets, $P_1(r\sin\alpha\cos\theta_1, r\sin\alpha\sin\theta_1, 0)$ and $P_2(r\sin\alpha\cos\theta_2, r\sin\alpha\sin\theta_2, 0)$, with the scattering coefficient A_{P1} and A_{P2}, respectively, (27) can be written as (29), as shown at the bottom of the next page.

According to (29), the two point targets, P_1 and P_2, with the left–right Doppler ambiguity have different chirp rates, namely $\gamma_2(\theta_1)$ and $\gamma_2(\theta_2)$. We need to separate the ambiguous targets and then execute MOCO on the ambiguous targets separately.

To solve the Doppler ambiguity, an advanced algorithm for left–right Doppler ambiguity via adaptive beamforming is proposed. As shown in Fig. 5, the mismatch of the space–time spectrum caused by the motion errors can be regarded as azimuth-invariant. Consequently, we can utilize spectral estimation to solve the mismatch of the space–time spectrum before beamforming. According to (29), the number of sources is 2 in each imaging unit. Let $R_X(\theta_1, \theta_2)$ demote the sample covariance matrix of the targets with azimuth angle θ_1 and θ_2. Eigenvalue decomposition of $R_X(\theta_1, \theta_2)$ is carried out to obtain signal subspace E_s and noise subspace E_n. Then, azimuth angles θ_1 and θ_2 of the ambiguous targets can be estimated by MUSIC [48], [49], that is

$$
P_{\text{MUSIC}}(\theta) = \frac{1}{V^H(\theta)E_N E_N^H V(\theta)}. \qquad (30)
$$

In the process of beamforming, we use the linearly constrained minimum variance (LCMV) algorithm [50]. The LCMV algorithm can be formulated as follows:

$$
\begin{cases}
\min_{\mathbf{W_n}} \mathbf{W}_n^H \mathbf{R}_X \mathbf{W}_n \\
\text{s.t.} \mathbf{W}_n^H \mathbf{V} = \mathbf{g}_n^H, \quad n = 1, \ldots, N
\end{cases} \qquad (31)
$$

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where $\mathbf{W}_n$ is the weight to be determined, the natural number N is the dimension of the left–right Doppler ambiguity in FLMC-SAR imaging, and $\mathbf{g}_n$ is the nth column of $N \times N$ identity matrix. Given the signal received by the array X, the sample covariance matrix R_X and the steering vector matrix $\mathbf{V} = [\,V(\theta_1)\ V(\theta_2)\,]^H$, the optimal weight of the beamforming filter is determined via the method of Lagrange multipliers

$$\mathbf{W}_n = \mathbf{R}^{-1}\mathbf{V}(\mathbf{V}^H\mathbf{R}^{-1}\mathbf{V})^{-1}\mathbf{g}_n. \tag{32}$$

With the optimal weight of the beamforming filter, beamforming is accomplished by weighting the ambiguity imaging result of each channel and accumulating them coherently to achieve imaging without Doppler ambiguity. For the left–right Doppler ambiguity problem of FLMC-SAR imaging, the dimension of the left-right Doppler ambiguity is 2. Let $\mathbf{W} = [\,\mathbf{W}_1{}^H\ \mathbf{W}_2{}^H\,]$, and then, the imaging result can be obtained as follows:

$$\begin{aligned}
&[\,s_4(\hat{t}, t_m; r, \theta_1)\ s_4(\hat{t}, t_m; r, \theta_2)\,] \\
&\quad = \left[\, s_3(\hat{t}, t_m; r, \theta_1, l)\cdot \mathbf{W}_1^H\ s_3(\hat{t}, t_m; r, \theta_1, l)\cdot \mathbf{W}_2^H \,\right] \\
&\quad = \operatorname{sinc}\left[B\left(\hat{t} - \frac{2(r-r_0)}{c}\right)\right]\cdot \exp\left[-j\frac{4\pi}{\lambda}(r-r_0)\right] \\
&\qquad \cdot \exp\left\{-j\frac{4\pi}{\lambda}[-\sin\alpha v(\cos\theta_1 - 1)t_m]\right\} \\
&\qquad \cdot \exp\{-j\varphi_{\Delta f_d} t_m\} \\
&\qquad \cdot \begin{bmatrix} A_{P1}\cdot \exp\left[-j(\gamma_2(\theta_1)+\gamma_3)\dfrac{4\pi}{\lambda}t_m^2\right] \\ \\ A_{P2}\cdot \exp\left[-j(\gamma_2(\theta_2)+\gamma_3)\dfrac{4\pi}{\lambda}t_m^2\right] \end{bmatrix}^T \\
&\varphi_{\Delta f_d} = \frac{4\pi}{\lambda}(-\sin\alpha\cos\theta\Delta v_{x0} - \sin\alpha\sin\theta\Delta v_{y0} + \cos\alpha\Delta v_{z0}) \\
&\gamma_2(\theta) = -\sin\alpha\sin\theta a_y \\
&\gamma_3 = v\Delta v_{x0}/r + \cos\alpha a_z.
\end{aligned} \tag{33}$$

D. Compensation of Angle-Dependent Azimuth-Variant Phase Errors

The Doppler ambiguity is solved via beamforming. We distinguish the left–right Doppler ambiguous target. Next, we will perform MOCO on the left and right scenes, respectively.

In each subaperture, the parametric motion errors model of the azimuth variable QPE model is established. The motion parameters are expressed as follows:

$$\begin{aligned}
\gamma_2(\theta) + \gamma_3 &= -\sin\alpha\sin\theta a_y + \cos\alpha a_z \\
&= \Upsilon_a \sin\theta + \Upsilon_b
\end{aligned} \tag{34}$$

where Υ_a is a linear coefficient used to describe the correlation between the motion error and the azimuth angle in each subaperture and Υ_b is the constant coefficient. In this article, a joint multichannel angle estimation-based signal quadratic decomposition method is proposed to estimate the two motion parameters, Υ_a and Υ_b.

The flowchart of the joint multichannel angle estimation-based signal quadratic decomposition method is shown in Fig. 6. First, strong targets are selected to proceed the signal quadratic decomposition, in a procedure similar to the conventional PGA. The signal is then modeled as a quadratic form under the assumption that there are M target sources in the selected range bin

$$\begin{aligned}
S &= \begin{bmatrix} A_{P_1}\cdot\exp[j f_d(\theta_1)t_m] \\ \vdots \\ A_{P_m}\cdot\exp[j f_d(\theta_m)t_m] \\ \vdots \\ A_{P_M}\cdot\exp[j f_d(\theta_M)t_m] \end{bmatrix}^T \cdot \begin{bmatrix} \exp\left[j\Upsilon(\theta_1)t_m^2\right] \\ \vdots \\ \exp\left[j\Upsilon(\theta_m)t_m^2\right] \\ \vdots \\ \exp\left[j\Upsilon(\theta_M)t_m^2\right] \end{bmatrix} \\
&= \sum_{\theta_m=\theta_1}^{\theta_M} A_{P_m}\cdot\exp[j f_d(\theta_m)t_m]\cdot\exp\left[j\Upsilon(\theta_m)t_m^2\right]
\end{aligned} \tag{35}$$

where $f_d(\theta_m)$ is the Doppler frequency coordinate of the target and $\Upsilon(\theta_m)$ is the chirp rate.

$$V(\theta) = \left[\exp\left(j\frac{2\pi}{\lambda}d\sin\alpha\sin(\theta)\right)\quad \exp\left(j\frac{2\pi}{\lambda}2d\sin\alpha\sin(\theta)\right)\cdots\quad \exp\left(j\frac{2\pi}{\lambda}Ld\sin\alpha\sin(\theta)\right)\right]. \tag{28}$$

$$\begin{aligned}
s_3(\hat{t}, t_m; r, \theta_1, l) &= \operatorname{sinc}\left[B\left(\hat{t} - \frac{2(r-r_0)}{c}\right)\right]\cdot\exp\left[-j\frac{4\pi}{\lambda}(r-r_0)\right] \\
&\quad \cdot\exp\left\{-j\frac{4\pi}{\lambda}[-\sin\alpha v(\cos\theta_1 - 1)t_m]\right\}\cdot\exp\{-j\varphi_{\Delta f_d}t_m\} \\
&\quad \cdot\left\{A_{P1}\cdot\exp\left[-j(\gamma_2(\theta_1)+\gamma_3)\frac{4\pi}{\lambda}t_m^2\right]\cdot\exp\left[j\frac{2\pi}{\lambda}ld\sin\alpha\sin(\theta_1)\right]\right. \\
&\quad \left. + A_{P2}\cdot\exp\left[-j(\gamma_2(\theta_2)+\gamma_3)\frac{4\pi}{\lambda}t_m^2\right]\cdot\exp\left[j\frac{2\pi}{\lambda}ld\sin\alpha\sin(\theta_2)\right]\right\} \\
\varphi_{\Delta f_d} &= \frac{4\pi}{\lambda}(-\sin\alpha\cos\theta\Delta v_{x0} - \sin\alpha\sin\theta\Delta v_{y0} + \cos\alpha\Delta v_{z0}) \\
\gamma_2(\theta) &= -\sin\alpha\sin\theta a_y \\
\gamma_3 &= v\Delta v_{x0}/r + \cos\alpha a_z
\end{aligned} \tag{29}$$

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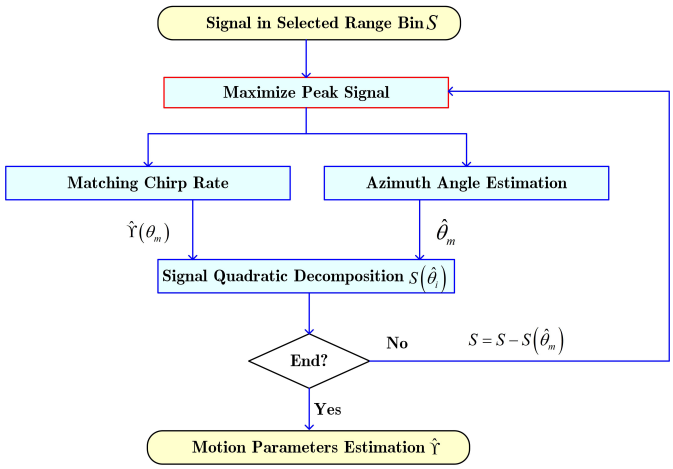

Fig. 6. Flowchart of the joint multichannel angle estimation-based signal quadratic decomposition method.

Second, the best matching chirp rate can be determined by (36)

$$\hat{\Upsilon}(\theta_m) = \arg_{\gamma} \max\left\{\max\left\{\mathrm{FFT}\left\{\exp\left[-j\gamma t_m^2\right]\cdot S\right\}\right\}\right\}. \quad (36)$$

Then, the azimuth angle θ_m of the target corresponding to the peak value in the signal can be determined by multichannel angle estimation as follows:

$$\hat{\theta}_m = \arg_{\theta_0} \max\{V^H(\theta_0)\cdot S_{\mathrm{peak}}\} \quad (37)$$

where θ_0 is the uniformly quantized azimuth angle grid and $V(\theta_0)$ is the corresponding steering vector. S_{peak} is the peak value as follows:

$$S_{\mathrm{peak}} = \max\left\{\mathrm{FFT}\left\{\exp\left[-j\hat{\Upsilon}(\theta_m)t_m^2\right]\cdot S\right\}\right\}. \quad (38)$$

Finally, the signal component of the target corresponding to the peak value in the signal can be determined

$$S(\hat{\theta}_m) = A_{P_m}\cdot\exp[j f_d(\hat{\theta}_m)t_m]\cdot\exp\left[j\hat{\Upsilon}(\hat{\theta}_m)t_m^2\right] \quad (39)$$

$$A_{P_m} = V^H(\hat{\theta}_m)\cdot S_{\mathrm{peak}}. \quad (40)$$

The target signal component $S(\hat{\theta}_m)$ is decomposed from the signal S. Iterating the above steps can achieve the signal quadratic decomposition of the signal S. Then, the azimuth angle $\hat{\theta}_m$ and the corresponding chirp rate $\hat{\Upsilon}(\hat{\theta}_m)$ are obtained.

Let $\Upsilon = [\Upsilon_a\ \Upsilon_b]$ be the motion parameters. According to the least square method, the motion parameters of the angle-dependent azimuth-variant error model can be obtained by solving the following minimization problem:

$$\hat{\Upsilon} = \arg_{\Upsilon}\min\|A\Upsilon - b\|_2^2 \quad (41)$$

where

$$\mathbf{A} = \begin{bmatrix} \sin\hat{\theta}_1 & 1 \\ \sin\hat{\theta}_2 & 1 \\ \vdots & \vdots \\ \sin\hat{\theta}_M & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} \hat{\Upsilon}(\hat{\theta}_1) \\ \hat{\Upsilon}(\hat{\theta}_2) \\ \vdots \\ \hat{\Upsilon}(\hat{\theta}_M) \end{bmatrix}$$

is the chirp rate estimate matrix. The motion parameters estimation of (41) is

$$\hat{\Upsilon} = (\mathrm{A}^{\mathrm{T}}\mathrm{A})^{-1}\mathrm{A}^{\mathrm{T}}\mathbf{b}. \quad (42)$$

With the estimated motion parameter $\hat{\Upsilon}$, the mapping relationship between azimuth angle θ and chirp rate $\gamma_2(\theta)$ at the center of each subaperture can be obtained. Then, the phase correction function to remove the angle-dependent azimuth-variant phase errors is given by

$$G_{\gamma 2}(\theta) = \exp\left(j\frac{4\pi}{\lambda}(\gamma_2(\theta)+\gamma_3)\left(t_c^k\right)^2\right) \quad (43)$$

where t_c^k is the slow time when the APC of the radar is located at the center of the kth subaperture. Then, the signals of left and right scene can be, respectively, expressed as

$$\begin{aligned} &s_5(\hat{t}, t_m; r, \theta_1) \\ &= \mathrm{IFFT}[\mathrm{FFT}[s_4(\hat{t}, t_m; r, \theta_1)] * G_{\gamma 2}(\theta_1)] \\ &= A_{P1}\cdot\mathrm{sinc}\left[B\left(\hat{t} - \frac{2(r-r_0)}{c}\right)\right]\cdot\exp\left[-j\frac{4\pi}{\lambda}(r-r_0)\right] \\ &\quad\cdot\exp\left\{-j\frac{4\pi}{\lambda}[-\sin\alpha v(\cos\theta_1 - 1)t_m]\right\}\cdot\exp\{-j\varphi_{\Delta f_d}t_m\} \\ &\varphi_{\Delta f_d} \\ &= \frac{4\pi}{\lambda}(-\sin\alpha\cos\theta\Delta v_{x0} - \sin\alpha\sin\theta\Delta v_{y0} + \cos\alpha\Delta v_{z0}) \end{aligned} \quad (44)$$

$$\begin{aligned} &s_5(\hat{t}, t_m; r, \theta_2) \\ &= \mathrm{IFFT}[\mathrm{FFT}[s_4(\hat{t}, t_m; r, \theta_2)] * G_{\gamma 2}(\theta_2)] \\ &= A_{P2}\cdot\mathrm{sinc}\left[B\left(\hat{t} - \frac{2(r-r_0)}{c}\right)\right]\cdot\exp\left[-j\frac{4\pi}{\lambda}(r-r_0)\right] \\ &\quad\cdot\exp\left\{-j\frac{4\pi}{\lambda}[-\sin\alpha v(\cos\theta_2 - 1)t_m]\right\}\cdot\exp\{-j\varphi_{\Delta f_d}t_m\} \\ &\varphi_{\Delta f_d} \\ &= \frac{4\pi}{\lambda}(-\sin\alpha\cos\theta\Delta v_{x0} - \sin\alpha\sin\theta\Delta v_{y0} + \cos\alpha\Delta v_{z0}). \end{aligned} \quad (45)$$

E. Full-Aperture Data Combination

Each of the subaperture data is a segment of a full-aperture data. In each subaperture, the primary angle-dependent phase errors are compensated. Then, the data of subapertures are coherently combined to recover the full-aperture data. There is a linear phase component between the adjacent subapertures, which should be removed before full-aperture data combination. The linear phase component between the adjacent subapertures can be estimated from the overlapped subapertures. The phases of the overlapping subapertures between the adjacent subapertures are subtracted to determine the linear phase component between the adjacent subapertures. To prevent discontinuity between the adjacent subapertures, the two adjacent subapertures should overlap in the coherent combination for obtaining the full aperture, as shown in Fig. 7.

The full-aperture data of left scene and right scene after MOCO can be, respectively, expressed as

$$\begin{aligned} s_6(\hat{t}, t_m; r, \theta_1) = {} & A_{P1}\cdot\mathrm{sinc}\left[B\left(\hat{t} - \frac{2(r-r_0)}{c}\right)\right] \\ & \cdot\exp\left[-j\frac{4\pi}{\lambda}(r-r_0)\right] \\ & \cdot\exp\left\{-j\frac{4\pi}{\lambda}[-\sin\alpha v(\cos\theta_1 - 1)t_m]\right\} \end{aligned}$$

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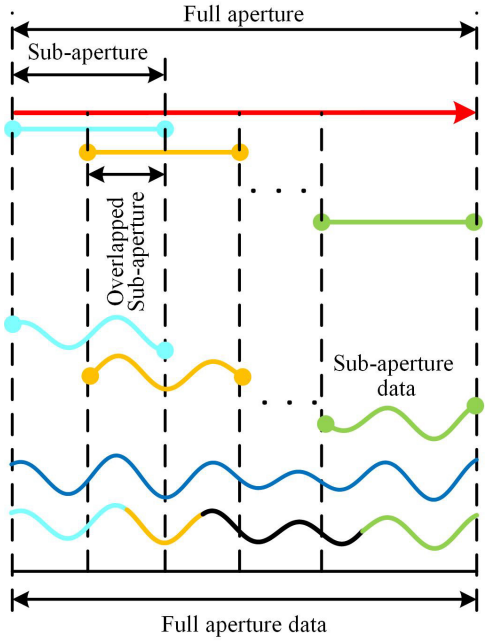

Fig. 7. Full-aperture data recovery.

$$-\frac{T_P}{2} \le t_m < \frac{T_P}{2} \tag{46}$$

$$\begin{aligned} s_6(\hat{t}, t_m; r, \theta_2) = {} & A_{P2} \cdot \operatorname{sinc}\left[B\left(\hat{t} - \frac{2(r-r_0)}{c}\right)\right] \\ & \cdot \exp\left[-j\frac{4\pi}{\lambda}(r-r_0)\right] \\ & \cdot \exp\left\{-j\frac{4\pi}{\lambda}[-\sin\alpha v(\cos\theta_2 - 1)t_m]\right\} \\ & -\frac{T_P}{2} \le t_m < \frac{T_P}{2}. \end{aligned} \tag{47}$$

Through the azimuth Fourier transform to the full-aperture data, the imaging results of FLMC-SAR can be yielded

$$\begin{aligned} & S_7(\hat{t}, t_m; r, f_d(\theta_1)) \\ & = \mathrm{FFT}[s_6(\hat{t}, t_m; r, \theta_1)] \\ & = 2\pi A_{P_1} \cdot \operatorname{sinc}\left[B\left(\hat{t} - \frac{2(r-r_0)}{c}\right)\right] \cdot \operatorname{sinc}[B_a(f - f_d(\theta_1))] \end{aligned} \tag{48}$$

$$\begin{aligned} & S_7(\hat{t}, t_m; r, f_d(\theta_2)) \\ & = \mathrm{FFT}[s_6(\hat{t}, t_m; r, \theta_2)] \\ & = 2\pi A_{P_2} \cdot \operatorname{sinc}\left[B\left(\hat{t} - \frac{2(r-r_0)}{c}\right)\right] \cdot \operatorname{sinc}[B_a(f - f_d(\theta_2))] \end{aligned} \tag{49}$$

where $f_d(\theta)$ is the Doppler frequency coordinate of the target with the beam pointing azimuth angle θ and B_a is the azimuth Doppler bandwidth. The flowchart of the proposed method is shown in Fig. 8.

V. Experiments and Discussion

In this section, we compare the simulation results of the proposed parametric azimuth-variant MOCO with those of the traditional autofocus methods, including PGA [35] and MD [47]. There are three cases in the experiment. Both the simulated and real-data experiments demonstrate that the proposed approach outperforms the conventional autofocus algorithms in terms of imaging quality.

TABLE I
Simulation Parameters

Parameter	Value	Parameter	Value
Carrier frequency	25GHz	PRF	8000Hz
Bandwidth	55MHz	Beam azimuth width	20degree
Number of array elements	5	Beam pitch width	22degree
Platform height	4000m	Real aperture length	0.04m
Platform velocity	1.02km/s	Number of channel	5

A. Case 1: Point Target Simulation

In this case, we give the point target simulation. As shown in Fig. 9, nine point targets are placed in the imaging area. Here, the X-axis denotes the target reference slant range and the Y-axis denotes the target azimuth angle. We define the portion of imaging area corresponding to an azimuth angle that is less than 0° as the left scene and define the rest of the imaging area is as the right scene. Thus, we can see five point targets in the left scene and four point targets in the right scene. We set the SNR to be 20 dB. A single-transmitter multiple-receiver array is adopted in the FLMC-SAR system, and the receiver is an array antenna with five channels. The relevant simulation parameters are shown in Table I. Motion errors in the Y- and Z-axis were added into the FLMC-SAR data. Motion errors in the X-axis are neglected here because they can be compensated by azimuth resample processing.

In the experiment, the simulation data contain 1024 pulses, and the full aperture is segmented into 16 subapertures. Each subaperture contains 64 pulses, and the overlapped subaperture between the adjacent subapertures contains 16 pulses. We compare the focusing results of different algorithms, including the proposed method, PGA, and MD. Fig. 10 shows the point targets' imaging results obtained through the proposed method. Affected by Doppler ambiguity, the targets that are symmetrical about the trajectory are aliased in the imaging results, as shown in Fig. 10(a). The procedure of Doppler ambiguity resolving is incorporated into the proposed algorithm, by which we can obtain the unambiguous imaging result. The unambiguous imaging results of the point targets in the left and right scenes are shown in Fig. 10(b) and (c), respectively.

The azimuth pulse response function of the point targets that are processed by the proposed method, conventional PGA, and MD, is shown in Fig. 11. There are nine point targets in the imaging area, five of them in the left scene and the rest in the right scene. As shown in Fig. 11, both PGA and MD fail to achieve MOCO for the left and right scenes due to the coupled problem of MOCO and Doppler ambiguity resolving, whereas the proposed method succeeds in this MOCO. In the case of point target imaging, PGA and MD cannot obtain unambiguous imaging results. On the other hand, the proposed

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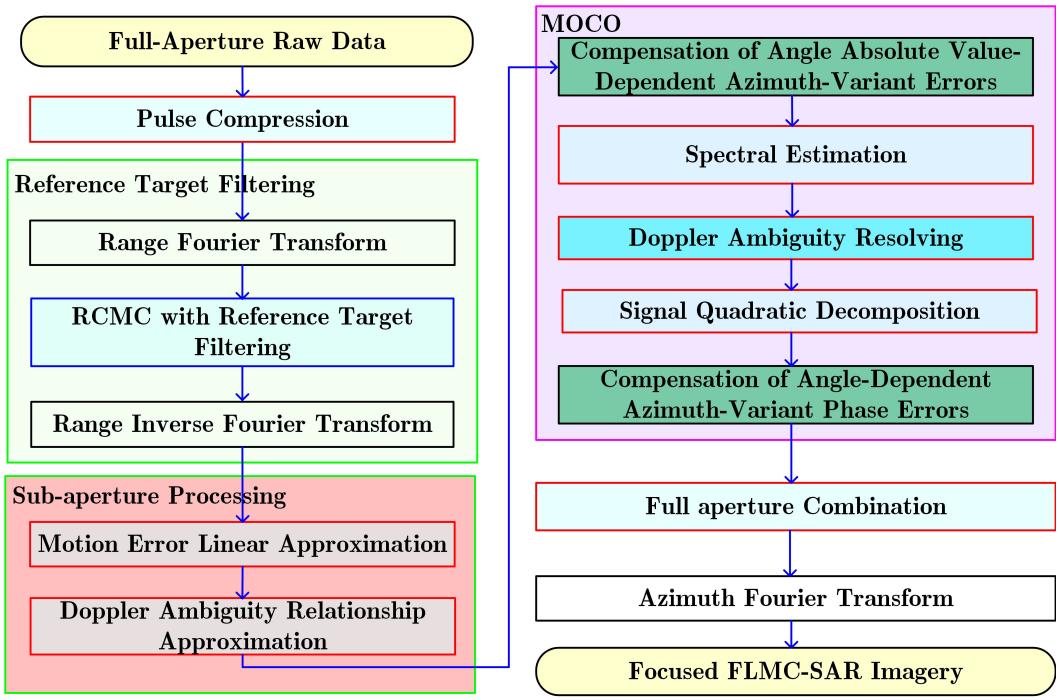

Fig. 8. Flowchart of the proposed method.

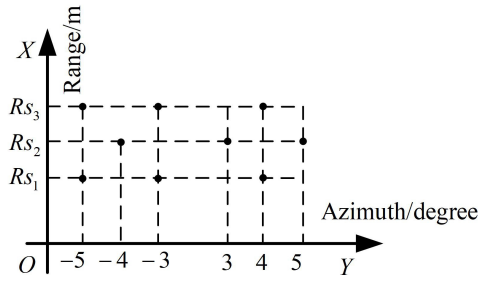

Fig. 9. Geometric distribution of the point targets.

method can greatly improve the focusing of the left and right scenes and achieve unambiguous imaging results. We compare the point impulse responses of the three algorithms in order to demonstrate the improvement of the proposed method on the focusing for the left and right scenes. There are three performance measurement metrics utilized to evaluate the point impulse responses, i.e., peak sidelobe ratio (PSLR), integrated sidelobe ratio (ISLR), and impulse response width (IRW), as shown in Table II. The values of PSLR, ISLR, and IRW obtained by the proposed method are lower than those obtained by both PGA and MD. It is demonstrated that the proposed method can improve the focusing for the left and right scenes. To show the robustness of the algorithm, the imaging results of point targets on different SNR conditions are also given. The azimuth pulse response function of the point target (Rs_2, 3°) on different SNR conditions is shown in Fig. 12. As the SNR increases, the PSLR, ISLR, and IRW of the imaging results decrease. Under the condition of 10-dB SNR, the proposed algorithm can still achieve the focused imaging results with 5.81-m IRW, −9.56-dB ISLR, and −13.04-dB PSLR.

Also, we give the performance measurement metrics (PSLR, ISLR, and IRW) of the point targets' imaging results with different subaperture lengths, as shown in Fig. 13. It can be seen from Fig. 13 that some aperture numbers are necessary for the proposed algorithm to ensure that the approximate conditions in the subapertures are satisfied.

B. Case 2: Surface Target Simulation

In this section, we performed the simulation experiment of a surface target. We set the SNR to be 20 dB, and the relevant simulation parameters are shown in Table I. The simulation data contain 1024 pulses, and the full aperture is segmented into 16 subapertures. Each subaperture contains 64 pulses, and the overlapped subaperture between the adjacent subapertures contains 16 pulses. To verify the effectiveness of the proposed algorithm for FLMC-SAR autofocus, we compare the imaging results of the proposed method to those of the PGA and MD. The image entropy was calculated as a performance measurement metric to evaluate the focusing performance of these algorithms. The reference image with entropy 12.87 is used to generate the simulation data, as shown in Fig. 14(a). Motion errors in the Y- and Z-axes are added into the FLMC-SAR data based on the geometry shown in Fig. 1 so that the reference image becomes blurred as shown in Fig. 14(b), whose entropy increases to 13.24. Then, we use MD, PGA, and the proposed algorithm to autofocus the blurred image. However, since MD and PGA cannot solve Doppler ambiguity, we first employ beamforming to resolve the Doppler ambiguity [26] before performing the image autofocusing through the two methods. The imaging results obtained by MD, PGA, and the proposed method are presented in Fig. 14(c)–(e), respectively. In addition, we use rectangles to highlight and magnify the two areas where artificial structures appear in each imaging result. In comparison to the proposed method, both MD and PGA exhibit poorer focusing results

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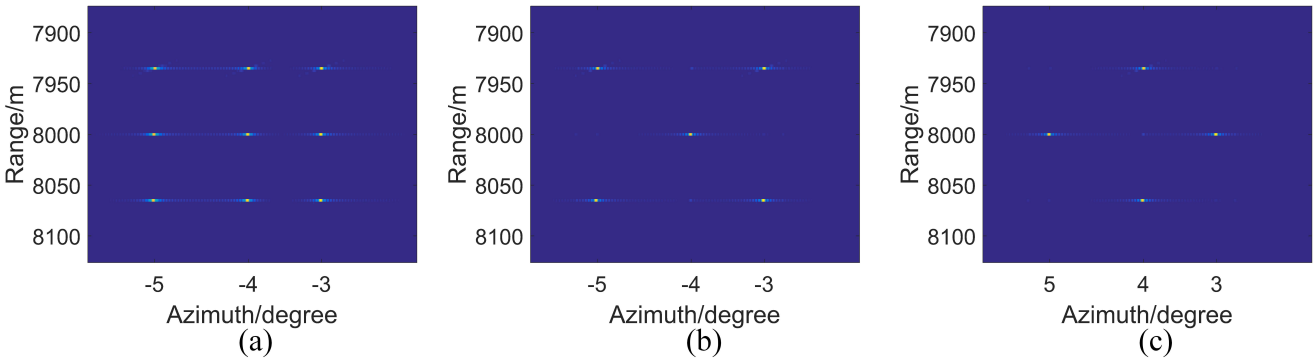

Fig. 10. Point targets' imaging results processed by the proposed method. (a) Ambiguous imaging results. (b) Ambiguity resolving result of the left scene. (c) Ambiguity resolving result of the right scene.

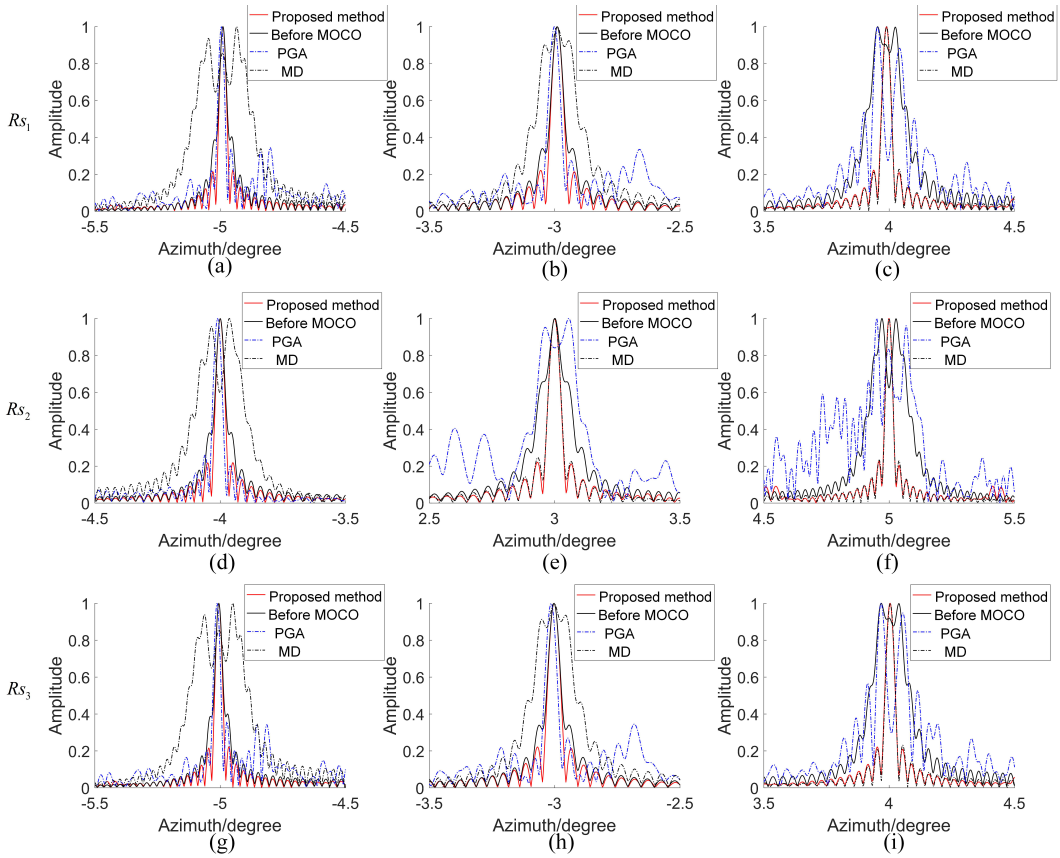

Fig. 11. Azimuth pulse response function of point targets processed by the proposed method, PGA, and MD. (a) Point target in Rs_1 range at azimuth angle $-5°$. (b) Point target in Rs_1 range at azimuth angle $-3°$. (c) Point target in Rs_1 range at azimuth angle $4°$. (d) Point target in Rs_2 range at azimuth angle $-4°$. (e) Point target in Rs_2 range at azimuth angle $3°$. (f) Point target in Rs_2 range at azimuth angle $5°$. (g) Point target in Rs_3 range at azimuth angle $-5°$. (h) Point target in Rs_3 range at azimuth angle $-3°$. (i) Point target in Rs_3 range at azimuth angle $4°$.

due to the effects of Doppler ambiguity on MOCO. The proposed method separates the coupled problems of Doppler ambiguity resolution and MOCO by subaperture division, thus significantly improving the image quality. With regard to a point-like target highlighted by the red circle, the proposed algorithm performs better than the other algorithms, as shown in Fig. 14(f). Therefore, compared with MD and PGA, the proposed algorithm can obtain better focused images with lower entropy. The entropy of the surface targets' imaging results on different SNR conditions was calculated as a performance measurement metric to evaluate the focusing performance of these algorithms, as shown in Fig. 15. As the SNR increases, the entropy of the imaging results processed by different algorithm decrease. Under the condition of 10-dB SNR, the proposed algorithm can also obtain better focused images with the 13.10 entropy.

We also give the entropy of the surface targets' imaging results with different subaperture lengths, as shown in Fig. 16. In the surface targets simulation experiment, some aperture numbers are also necessary for the proposed algorithm to

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TABLE II
PSLR, ISLR, AND IRW OBTAINED BY THE PROPOSED METHOD, PGA, AND MD

	Azimuth angle	Reference slant range	Proposed method			PGA			MD		
			PSLR/dB	ISLR/dB	IRW/m	PSLR/dB	ISLR/dB	IRW/m	PSLR/dB	ISLR/dB	IRW/m
Left scene	-5	Rs_1	-13.02	-9.84	3.45	-13.02	-9.84	3.46	-0.57	5.38	8.65
	-3	Rs_1	-13.11	-9.89	5.80	-11.77	-8.34	5.80	-0.68	2.73	19.37
	-4	Rs_2	-13.18	-9.88	4.33	-11.69	-8.39	4.33	-0.38	0.27	10.05
	-5	Rs_3	-13.02	-9.79	3.46	-8.94	-6.82	3.46	-0.55	5.46	8.73
	-3	Rs_3	-13.08	-9.92	5.80	-11.39	-7.37	5.81	-0.58	3.12	19.79
Right scene	4	Rs_1	-13.04	-9.51	4.34	-0.02	1.73	15.82	-12.83	-9.50	4.34
	3	Rs_2	-13.09	-9.67	5.80	-0.42	-0.81	21.01	-12.15	-9.26	5.81
	5	Rs_2	-12.94	-9.35	3.46	-5.74	-1.23	13.46	-12.61	-9.25	3.46
	4	Rs_3	-13.11	-9.57	4.33	-0.47	3.18	16.56	-12.70	-9.55	4.34

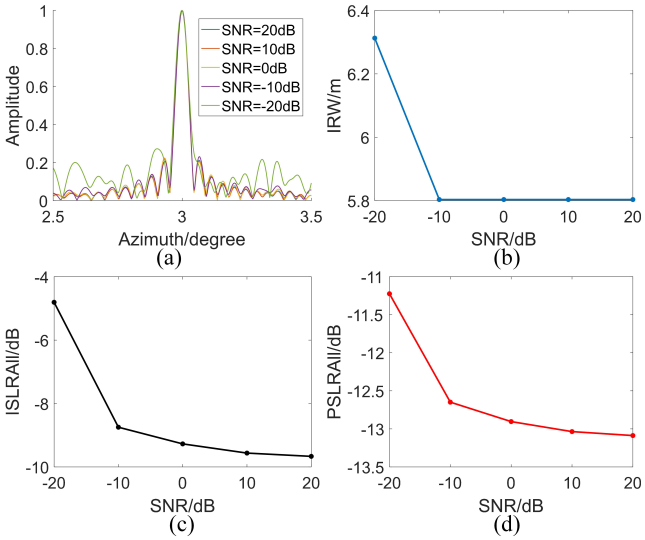

Fig. 12. Imaging result of the point target (Rs_2, 3°) on different SNR conditions. (a) Azimuth pulse response function of the point target on different SNR conditions. (b) IRW of the point target on different SNR conditions. (c) ISLR of the point target on different SNR conditions. (d) PSLR of the point target on different SNR conditions.

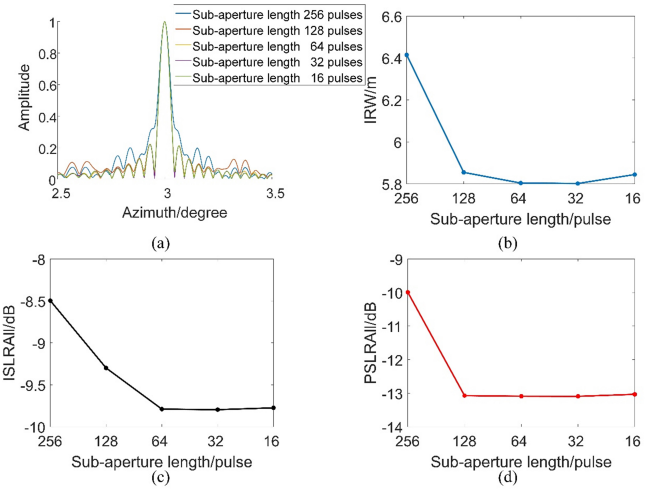

Fig. 13. Imaging result of the point target (Rs_2, 3°) with different subaperture lengths. (a) Azimuth pulse response function of the point target with different subaperture lengths. (b) IRW of the point target with different subaperture lengths. (c) ISLR of the point target with different subaperture lengths. (d) PSLR of the point target with different subaperture lengths.

ensure that the approximate conditions in the subapertures are satisfied.

C. Case 3: Real Data Experiment

In this section, the real data experiment is carried out to demonstrate the performance of the proposed method on FLMC-SAR imaging. The real data are collected by a K-band experimental FLMC-SAR radar equipped with a five-channel array antenna. The pulse repetition frequency (PRF) is 6000 Hz, the synthetic aperture time is about 1.3 s containing 8192 pulses, and the full aperture is segmented into 128 subapertures. Each subaperture contains 64 pulses, and the overlapped subaperture between the adjacent subapertures contains 16 pulses. The radar is mounted on an aircraft platform at the altitude of 4000 m and a speed of 80 m/s. The reference slant range of the center of the scene is 8000 m, with 8° beam azimuth width and 20° beam pitch width. Due to the limitation of the INS/GPS, the motion information from the INS/GPS is not sufficient for highly precise MOCO. Therefore, MOCO based on the raw data is necessary for FLMC-SAR imaging. The imaging result obtained by INS/GPS-based MOCO, PGA, and the proposed method is shown in Fig. 17(a)–(c), respectively. The satellite image is shown in Fig. 17(d), and the red line is the trajectory. The resolution of the imaging area near the trajectory is limited in this system [27]. The image is projected in geographic coordinates. The vertical direction is latitude, and the horizontal direction is longitude. Obviously, due to the inaccuracy of the motion error information, the image generated by INS/GPS-based MOCO is severely defocused. The typical areas having artificial structures, named Scene A, are highlighted by red rectangles in Fig. 17(b)–(d). They are magnified as shown in Fig. 17(e)–(g), respectively. Obviously, the image processed by the proposed algorithm has a better focusing effect. In Fig. 17(b)–(d), we use yellow circles to highlight the seriously blurred areas, which are named Scene B and, respectively, magnified as shown in Fig. 17(h)–(j). It can be seen that the roads and buildings cannot be distinguished via the traditional PGA. A point-like target is highlighted by the red circle in Scene B, and its azimuth response function is shown in Fig. 17(k).

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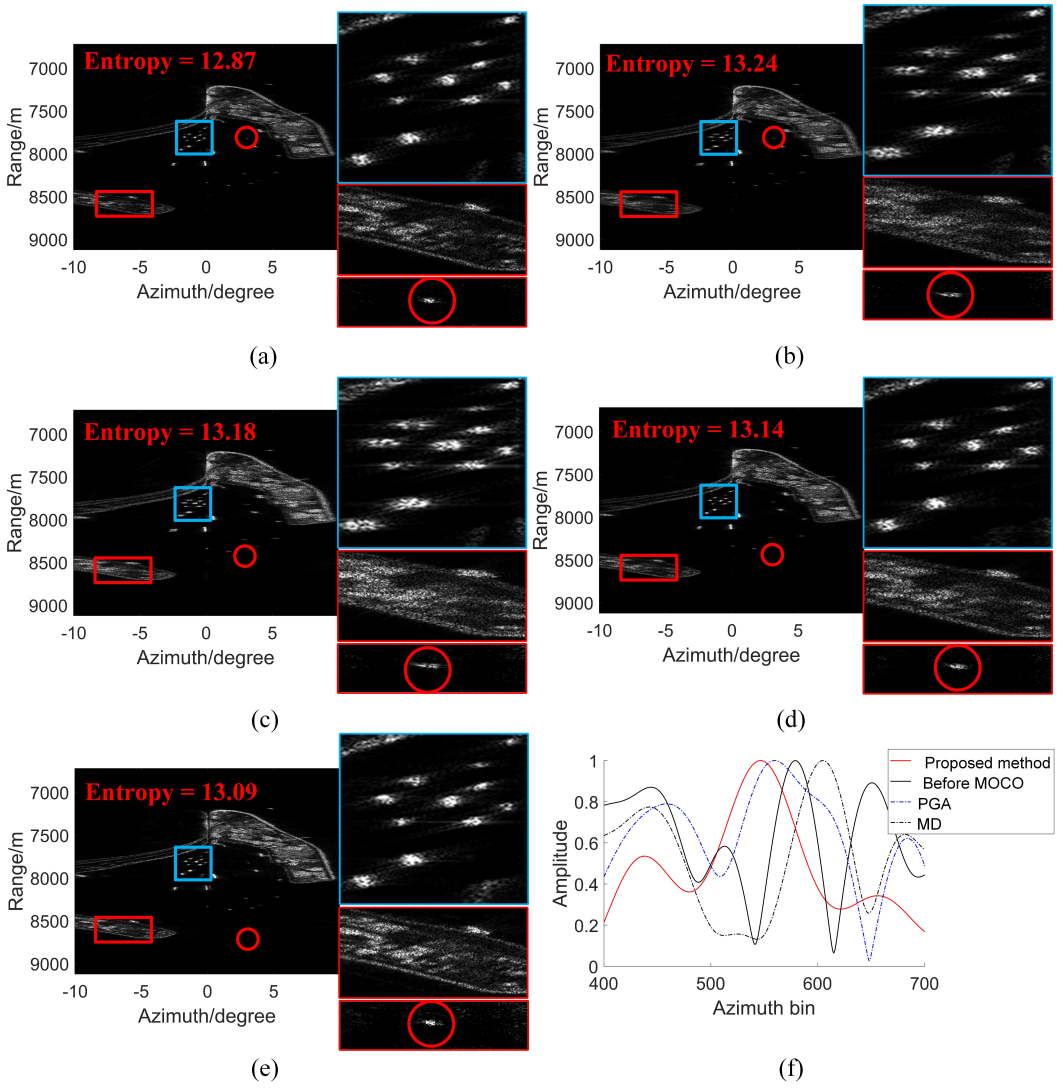

Fig. 14. Surface targets' imaging results. (a) Reference imaging results without motion errors. (b) Defocused imaging results without MOCO. (c) Imaging results processed by MD. (d) Imaging results processed by PGA. (e) Imaging results processed by the proposed method. (f) Comparison of azimuth pulse response function processed by different algorithms.

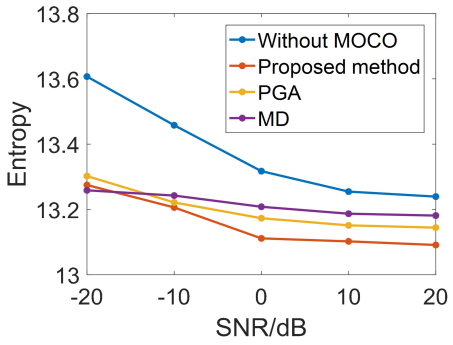

Fig. 15. Entropy of the surface targets' imaging results on different SNR conditions.

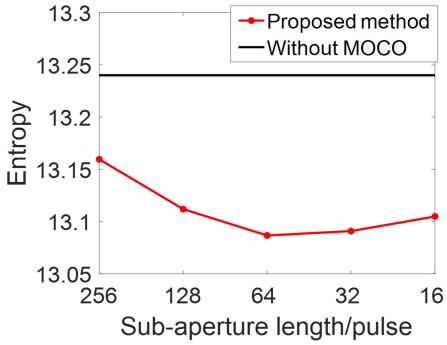

Fig. 16. Entropy of the surface targets' imaging results with different subaperture lengths.

Compared with other algorithms, the proposed algorithm achieves well-focused imaging. Due to the coupling problem between Doppler ambiguity resolving and MOCO, PGA shows poor focusing effect. Through the subaperture division, the proposed method achieves the decoupling of Doppler ambiguity resolving and MOCO, thereby significantly improving image quality. With the proposed MOCO method, the imaging result is well-focused, and the roads and buildings can be distinguished from each other. The real data

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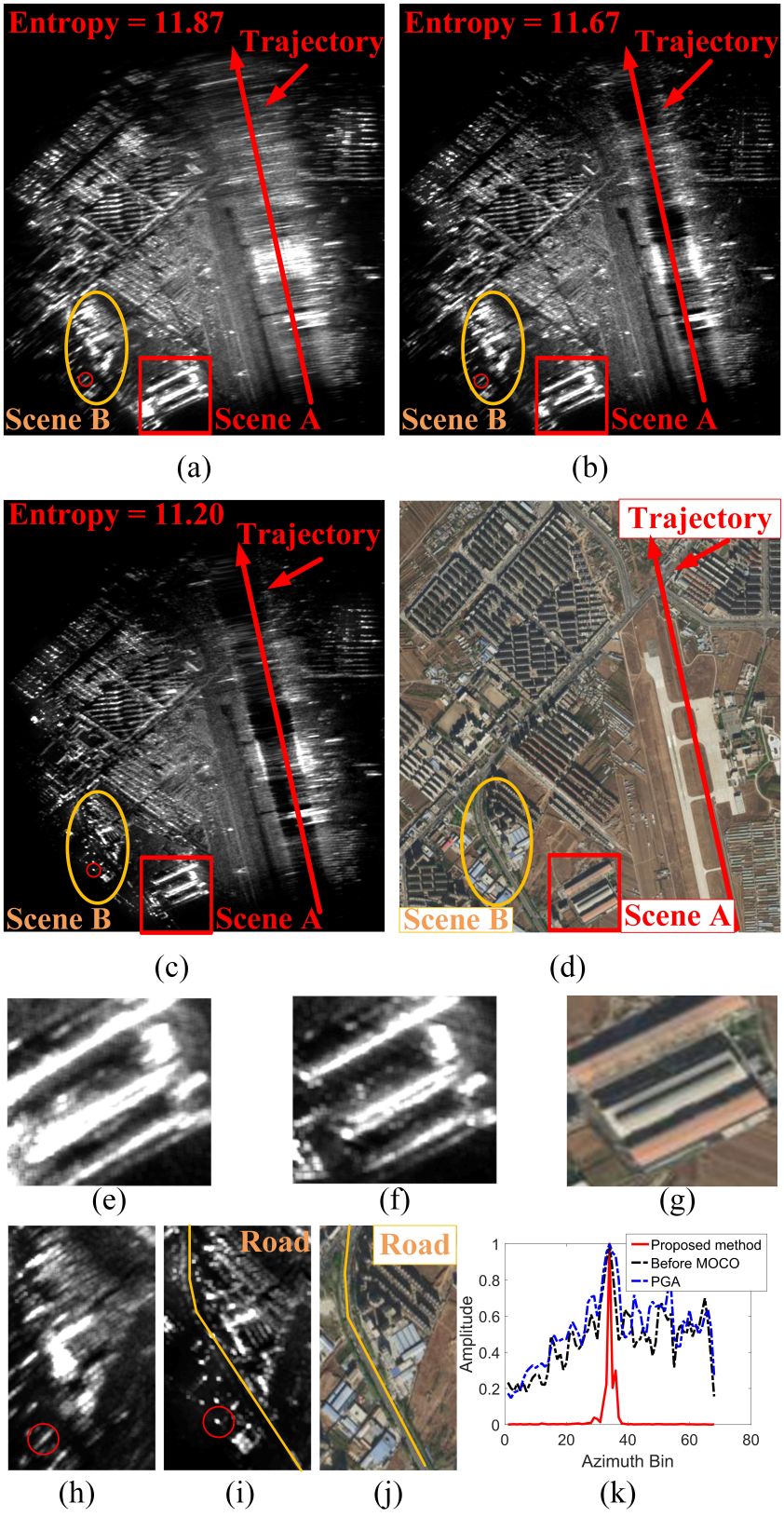

Fig. 17. FLMC-SAR images of real data. (a) Image processed by INS/GPS-based MOCO. (b) Image processed by traditional PGA. (c) Image processed by the proposed MOCO. (d) Satellite image of imaging area. (e) Magnified images of Scene A with traditional PGA. (f) Magnified images of Scene A with the proposed algorithm. (g) Magnified satellite image of Scene A. (h) Magnified images of Scene B with traditional PGA. (i) Magnified images of Scene B with the proposed algorithm. (j) Magnified satellite image of Scene B. (k) Comparison of azimuth pulse response function processed by different algorithms.

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experiment verifies the effectiveness of the proposed algorithm for FLMC-SAR imaging.

VI. Conclusion

FLMC-SAR is an important tool for modern remote sensing applications, which has the capability to reconstruct the high-resolution image of the front area. However, due to the azimuth-variant characteristics of motion errors over a long aperture, FLMC-SAR data processing is usually a challenging task. To accomplish an accurate MOCO for FLMC-SAR, a novel parametric azimuth-variant MOCO approach is proposed in this article. Since the motion errors are approximately invariant within a short azimuth aperture, we decompose a full synthetic aperture into several subapertures so that the high-order motion errors of the full aperture is decomposed into the first-order motion errors of the subapertures. Then, the phase errors are modeled as two terms in each subaperture: angle absolute value-dependent azimuth-variant phase errors and angle-dependent azimuth-variant phase errors. A joint multichannel angle estimation-based signal quadratic decomposition method is proposed to estimate the motion parameters. Before Doppler ambiguity resolving, the angle absolute value-dependent azimuth-variant phase errors for the left and right scenes can be compensated, but the angle-dependent azimuth-variant phase errors should be compensated individually for the left and right scenes after Doppler ambiguity resolving. Finally, the subaperture data are coherently combined to recover the full-aperture data. Experimental results based on both simulated and real data demonstrate that the proposed approach outperforms the conventional autofocus algorithms in terms of imaging quality.

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